Chapter 1.7 - Group Actions

Emil Marinov

February 25, 2026

1 Problem 20

Let G be a group and let G act upon itself by left conjugation, so each $g \in G$ maps G to G by

$$x \to gxg^{-1} \tag{1}$$

*For fixed $g \in G$, prove that conjugation by g is an isomorphism from G onto itself (i.e., is an **automorphism** of G - cf. Exercise 20, Section 6). Deduce that x and gxg^{-1} have the same order for all x in G and that for any subset A of G, $|A| = |gAg^{-1}|$ (here $gAg^{-1}|a \in A$).*

To show that the left conjugation on G is an isomorphism, it must be shown that 1) it is a homomorphism, 2) it is injective, and 3) it is surjective.

Show Homomorphic

$$\begin{aligned} \varphi(xy) &= g(xy)g^{-1} \\ &= gxyg^{-1} \quad \text{By associativity.} \\ &= gxg^{-1}gyg^{-1} \\ &= (gxg^{-1})(gyg^{-1}) \\ &= \varphi(x)\varphi(y) \end{aligned} \tag{2}$$

Show Injective

Here, it is needed to be shown that $\varphi(x) = \varphi(y) \implies x = y$.

$$\begin{aligned} \varphi(x) = gxg^{-1} = gyg^{-1} &= \varphi(y) \\ gxg^{-1}g &= gyg^{-1}g \\ gx &= gy \\ g^{-1}gx &= g^{-1}gy \\ x &= y \end{aligned} \tag{3}$$

Show Surjective

To show surjectivity, it must be shown that each element from the codomain has an element in the domain that maps to it: $\forall y \in G, \exists x \in G \quad \text{s.t.} \varphi(x) = y$.

$$\begin{aligned} y = \varphi(x) &= gxg^{-1} \\ \implies x &= g^{-1}yg \end{aligned} \tag{4}$$

This holds for all $g \in G$. Thus, it is surjective.

Order

Let n be the order of x, i.e. $x^n = 1$. Let m be the order of $\varphi(x)$, i.e. $\varphi(x)^m = 1$.

$$\begin{aligned} \varphi(x)^m = (gxg^{-1})^m = gx^m g^{-1} &= e \\ x^m = g^{-1}eg = g^{-1}g &= e \end{aligned} \tag{5}$$

If $x^n = e$, then $gx^n g^{-1} = e$. This implies $|\varphi(x)| = m \leq n$. If $gx^m g^{-1} = e$, then $x^m = e$. This implies $|x| = n \leq m$.

Because $m \leq n$ and $n \leq m$, $m = n$. Thus, the order of the automorphism is the same as that for the input group element.

Order 2

gAg^{-1} is bijective with A. Thus, $|A| = |gAg^{-1}|$.